FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Main)-2021

TEST DATE: 04-01-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. $x_f - x = x_f e^{-\lambda t}$

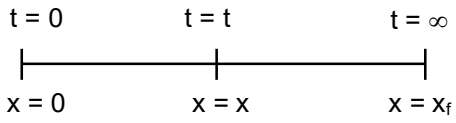
$\lambda = \dfrac{\ell n2}{T_{half}}$

$x = x_f\left(1 - e^{-\lambda t}\right)$

$\dfrac{dx}{dt} = x_f \lambda e^{-\lambda t}$

$V = x_f \lambda e^{-\lambda t}$ at t = 0, V = 10 m/s

$\Rightarrow \quad x_f = \dfrac{10}{\lambda} = \dfrac{10}{\ell n2}$



2. B

Sol. $y = x\tan\theta - \dfrac{gx^2}{2u^2\cos^2\theta}$

$\Rightarrow \quad gx^2\sec^2\theta - 2u^2 x\tan\theta + 2u^2 y = 0$

$\Rightarrow \quad gx^2\tan^2\theta - 2u^2 x\tan\theta + 2u^2 y + gx^2 = 0$

For real roots of $\tan\theta$

$4u^4x^2 - 4\left(2u^2y + gx^2\right)\left(gx^2\right) \geq 0$

$\Rightarrow \quad y \leq \dfrac{u^4 - g^2x^2}{2u^2g}$

3. C
Sol. Self-explanatory

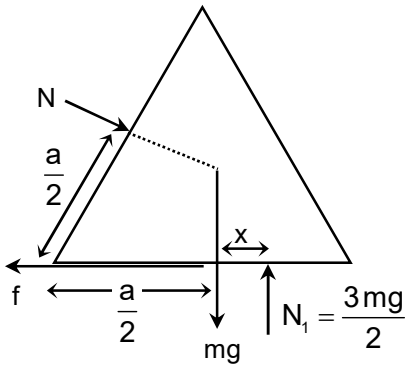
4. C
Sol. Torque equation about left corner –

$$N\frac{a}{2}+mg\frac{a}{2}=N_1\left(\frac{a}{2}+x\right)$$

$$mg\frac{a}{2}+mg\frac{a}{2}=\frac{3mg}{2}\left(\frac{a}{2}+x\right)$$

$$a=\frac{3}{4}a+\frac{3}{2}x$$

$$\frac{a}{6}=x$$



5. C
Sol. Pressure difference on an element at radial distance r and depth h = $(d\rho)gh$
Net force on element

$$\Rightarrow \quad (dA)((d\rho)gh)=(dm)\omega^2 r$$

$$\Rightarrow \quad (dA)((d\rho)gh)=(\rho(dA)dr)\omega^2 r$$

$$\Rightarrow \quad \frac{d\rho}{\rho}=\frac{\omega^2 r dr}{gh}$$

$$\Rightarrow \quad \log\frac{\rho}{\rho_0}=\frac{\omega^2 r^2}{2gh}$$

$$\rho=\rho_0 e^{\omega^2 r^2/2gh}$$

6. B
Sol.

$$U\alpha\frac{1}{\rho}$$

$$\Rightarrow \frac{f}{2}nRT\alpha\frac{V}{M}$$

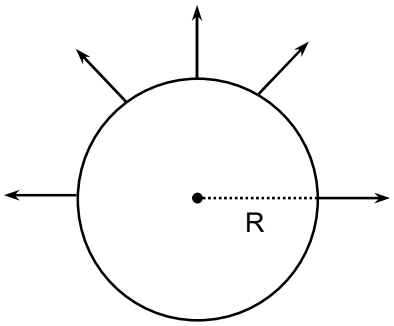
$$\Rightarrow T\alpha V$$

7. D
Sol. In equilibrium of upper half of loop
2kx = 4RS

$$x=\frac{2SR}{k}$$

$$2\pi R=\ell+\frac{2SR}{k}$$

$$R=\frac{k\ell}{2\pi k-2S}$$



8. A
Sol. Let natural radius of loop is R and at any instant radius is r then mechanical energy of loop is

$$\frac{1}{2}K\left(2\pi r-2\pi R\right)^2+\frac{1}{2}Mv^2=\text{constant}$$

$$K2\pi\left(r-R\right)2\pi+\frac{Mvdv}{dr}=0$$

$$a=-4\pi^2\frac{K}{M}\left(r-R\right)$$

$$T=\sqrt{\frac{M}{K}}$$

9. C

Sol. $\frac{\ell_2-3\ell_1}{2}=e$

Absolute error will be same in all and least percentage error is when e is maximum.

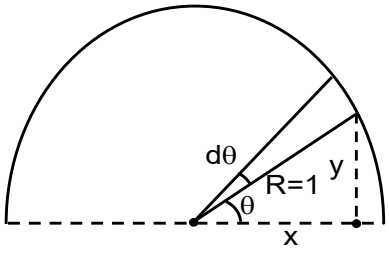
10. A

Sol. $x=\cos\theta$

$y=\sin\theta$

$dq=\lambda d\theta$

$dU=Vdq$



$$=\left(4\cos^2\theta-1\right)\sin\theta\lambda d\theta$$

$$=\lambda\left(4\left(1-\sin^2\theta\right)-1\right)\sin\theta d\theta$$

$$=\lambda\left(3\sin\theta-4\sin^3\theta\right)d\theta$$

$$=\int\lambda\sin3\theta d\theta$$

$$U=\lambda\left(\frac{-\cos3\theta}{3}\right)_0^{\pi}=\frac{2}{3}\lambda$$

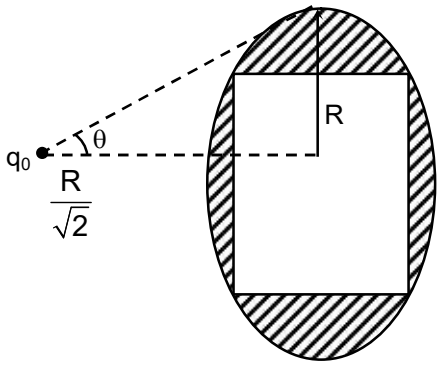
11. B

Sol. $\phi=\phi_{disc}-\phi_{square}$

$$=\frac{q}{2\varepsilon_0}\left(1-\cos\theta\right)-\frac{q}{6\varepsilon_0}$$

$$=\frac{q}{2\varepsilon_0}\left(1-\frac{1}{\sqrt{3}}-\frac{1}{6}\right)$$

$$=\frac{q}{12\varepsilon_0}\left(5-2\sqrt{3}\right)$$



12. C

Sol. Total power in horizontal resistors at any time = $i^2R+\frac{i^2}{4}R+\frac{i^2}{16}R+.....\infty$

$=\frac{4}{3}i^2R$

Total power in vertical resistors =

$$= \frac{i^2}{4}2R + \frac{i^2}{16}2R + ...\infty$$

$$= \frac{2}{3}i^2R$$

$$\frac{P_{Horizontal}}{P_{Vertical}} = 2$$

13. B

Sol. In middle region of magnetic field particle moves in a circle centered at C then

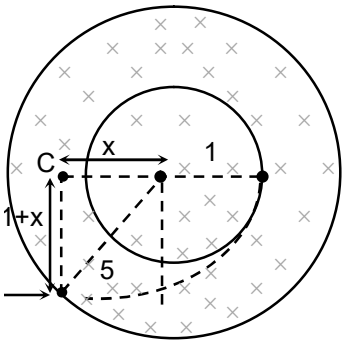
$5^2 = x^2 + (1+x)^2$

$x = 3m$

Radius of circular

Path in middle region = 4m

So $\frac{B_1}{B_2} = \frac{r_2}{r_1} = \frac{4}{1}$



14. D

Sol. During motion of rod

$$B\ell v = \frac{Ldi}{dt}$$

$$\Rightarrow \quad B\ell x = Li$$

Energy stored in inductor

$$= \frac{1}{2}Li^2 = \frac{B^2\ell^2x^2}{2L}$$

Energy lost = $K - \frac{B^2\ell^2x^2}{2L}$

15. C

Sol. Initially

$$\varepsilon = \frac{Ldi}{dt}$$

$Li = \varepsilon t$

Energy stored in inductor at $t = t_0$

$$= \frac{\varepsilon^2 t_0^2}{2L}$$

Which is also total energy lost.

16. A

Sol. Shifting of lens is

= 6 × 1 = 6 cm

u = – x

v = x + 6

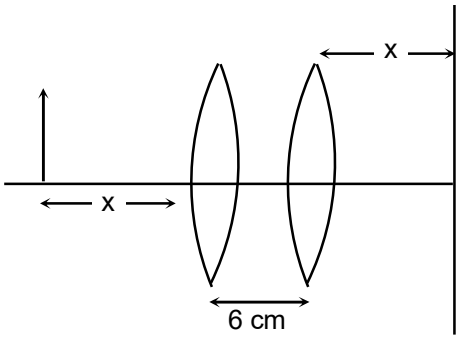
$$\frac{x+6}{-x} = -2$$

$\Rightarrow$ x = 6 cm

u = – 6 cm

v = 12 cm

$$\Rightarrow \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{12} + \frac{1}{6} = \frac{1}{f} \Rightarrow f = 4\,cm$$



17. B

Sol. Self-Explanatory

18. C

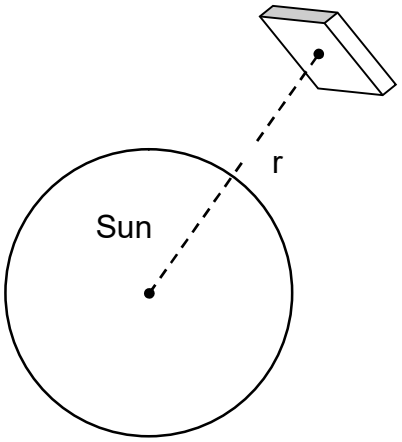
Sol. In equilibrium

$F_{gravity} = F_{Radiation}$

$$\Rightarrow \quad \frac{GM_s m_{plate}}{r^2} = \left(\frac{P}{4\pi r^2}\right)\frac{1}{c}A$$

$$\Rightarrow \quad \frac{GM_S}{r^2}\rho At = \frac{PA}{4\pi r^2 c}$$

$$\Rightarrow \quad t = \frac{P}{4\pi GM_S \rho c}$$



19. D

Sol. Practical diode cannot have zero resistance.

20. B

Sol. Momentum conservation

$$p_e^2 + \left(\frac{h}{2\lambda}\right)^2 = \left(\frac{h}{\lambda}\right)^2$$

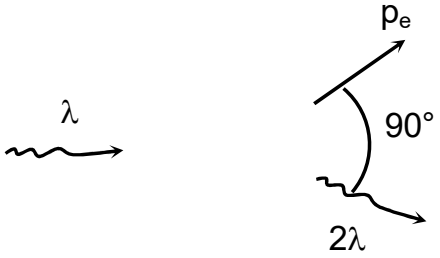
$$p_e^2 = \frac{3}{4}\frac{h^2}{\lambda^2}$$

Energy conservation

$$\frac{hc}{\lambda} = \frac{hc}{2\lambda} + \frac{p_e^2}{2M_e}$$

$$\Rightarrow \quad \frac{hc}{2\lambda} = \frac{p_e^2}{2M_e} = \left(\frac{3}{4}\frac{h^2}{\lambda^2}\right)\frac{1}{2M_e}$$

$$\Rightarrow \quad \lambda = \frac{3h}{4M_e c}$$



SECTION – B

21. 3

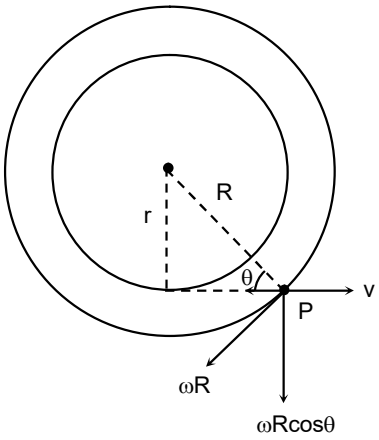
Sol. Horizontal velocity of point

$= v - \omega R\sin\theta$

$= \omega r - \omega R\left(\frac{r}{R}\right)$

$= 0$

So velocity must be perpendicular.



22. 2

Sol. $X_{CM} = \frac{2+8}{2} = 5m$

$d = X - X_{CM} = 3m$

$I = I_{CM} + Md^2$

$\Rightarrow 92 = I_{CM} + 10(3)^2$

$\Rightarrow I_{CM} = 2\ kgm^2$

SECTION – C

23. 00008.00

Sol. Energy of sinusoidal pulse $= \frac{1}{2}\mu\lambda\omega^2A^2 = \frac{1}{2}.\frac{T}{v^2}.\frac{v}{f}.\omega^2A^2$

$$\frac{E_{Transmitted}}{E_{Reflected}} = \frac{v_1A^2_{Transmitted}}{v_2A^2_{Reflected}} = \frac{1}{2}.\left(\frac{4}{1}\right)^2 = 8$$

24. 00003.20

Sol. Resistance between sphere and infinite $= \int_R^\infty \frac{\rho dr}{4\pi r^2} = \frac{\rho}{4\pi R}$

Capacitance of sphere $= 4\pi\varepsilon R$

Time constant $= \rho\varepsilon$

25. 00012.50

Sol. $\frac{P_S}{P_P} = \frac{v_si_s}{v_pi_p} = 0.8$

$\Rightarrow \frac{v_s.v_s/10}{v_p.i_p} = 0.8$

$\because v_s = \frac{v_p}{2}$

$\Rightarrow \frac{v_p^2}{40v_pi_p} = 0.8 \quad \Rightarrow i_p = \frac{v_p}{32} = 12.5\,A$

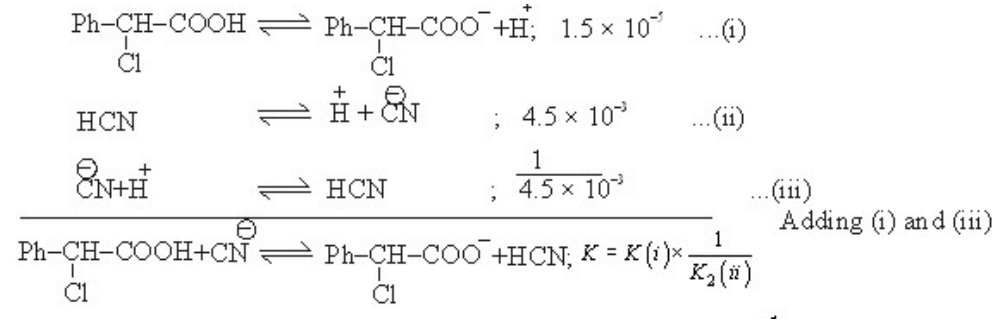
Chemistry

PART – II

SECTION – A

26. C

Sol.



$$= 1.5 \times 10^{-5} \times \frac{1}{4.5 \times 10^{-3}} = 3.3 \times 10^{-3}$$

27. D

Sol. Conceptual

28. D

Sol. Cl shows oxidation and reduction in HCl and $HClO_3$

29. B

Sol.

30. A

31. D

Sol. $PbS + 4H_2O_2 \rightarrow PbSO_4 +$

$= 0.4$ moles

32. B

Sol. The crystal of PBr_5 is found to contain $[PBr_4^+]$ and $[Br^-]$

33. C

Sol. $2H_2O \longrightarrow 2H_2(g) + O_2(g)$
$2x \quad x$

$3x = 168$

$x = 56$ ml

$V_{H_2} = 2x = 112$ ml

$V_{O_2} = x = 56$ ml

11200 ml of H_2 at STP = 1 F

112 ml of H_2 at STP = 0.01 F

Now

5600 ml of O_2 at STP = 1F

56 ml of O_2 at STP = 0.01 F

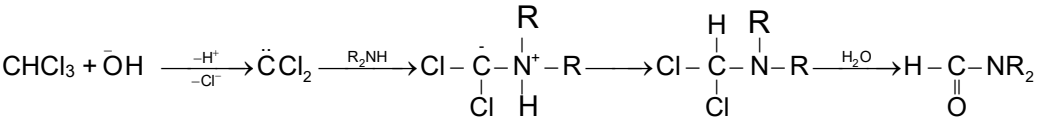
Amount of electricity passed = 0.01 F = $0.01 \times 96500 = 965$ C

We get identical results whether are consider H_2 or O_2

34. B

Sol. Carbylamine test is given by 1^0 amine only

Where as 2^0 amine reacts as follows:



35. B

Sol. $250x + 500y = 1.6 \times 2000$

$\frac{x}{y} = \frac{5}{4}$ ..(

By solving x = 4.92, y = 3.94

x + y = 8.86

36. C

37. D

Sol. Compounds containing $C, S \ \& \ N$ gives blood red colour in Lassaigne's test

38. C

Sol. $C_xH_{y(g)} + \left(x + \frac{y}{4}\right)O_{2(g)} \longrightarrow xCO_{2(g)} + y/_2\, H_2O(\ell)$

Initial volume	5	30	0	0
Final volume	0	$30 - \left(x + \frac{y}{4}\right)5$	$5x$	0

Volume of CO_2 absorbed = 10 ml

Volume of O_2 absorbed = 15 ml

$5x = 10$, $x = 2$

$30 - \left(x + \frac{y}{4}\right)5 = 15,\ y = 4 \quad \text{formula} = C_2H_4$

39. B

Sol. $+M$ effect of N

40. C

41. B

42. B

Sol. $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$

at 127^0C the no of gaseous mole will remain same because H_2O will be in vapour form.

These for the pressure change will be due to change in temperature

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \qquad \frac{1}{300K} = \frac{P}{400K}$$

$$P = \frac{400}{300} = \frac{4}{3} = 1.33\,atm$$

43. A

Sol. Concentration of $|H^+|$ at anode = 10^{-8} M

$\therefore$ concentration of $|H^+|$ at cathode = 10^{-7} M

because pH = $\frac{1}{2}\left[pkw + pka - pkb\right] = \frac{1}{2}[14 + 4.74 - 4.74] = 7.$

so $|H^+| = 10^{-7}$ M

E = 0 - $\frac{0.059}{1}\log\frac{10^{-8}}{10^{-7}}$ = E = 0.059 V

44. A

Sol. Suppose weight of each layer is w.

Total weight of phenol = $w \times 0.7 + w \times 0.1 = 0.8\,w$

Total weight of $H_2O = w \times 0.30 + w \times 0.90 = 1.20\,w$

$$\frac{weight\ of\ phenol}{Weight\ of\ H_2O} = \frac{x}{y} = \frac{0.8w}{1.20w} = \frac{2}{3}$$

45. D

Sol. I : BrF_4^+ : Trigonal bipyramidal structure and see saw shape

II : Bond order of O_2 increases by the removal of e^-, as bond order increases

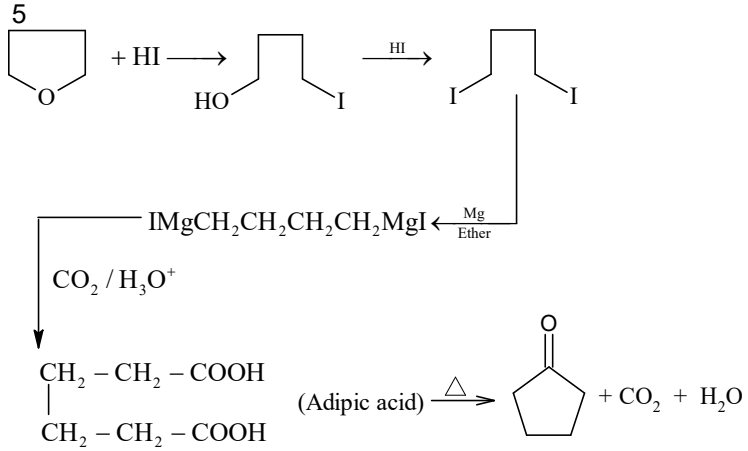
III : Br_3^- and Br_3^+ are of different shape and structure.

IV : In PCl_5, hybrid orbital of P is $sp^3d_{z^2}$

SECTION – B

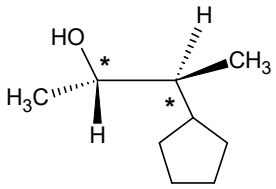
46. 5

Sol.



47. 2

Sol.



SECTION – C

48. 00001.41

Sol. At anode:

$H_2(g) \longrightarrow 2H^+(aq) + 2e^-, \quad E^\circ = 0\ V$

At cathode:

$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O, \quad E^\circ = 1.5\ V$

$E_{cell} = E^\circ_{cell} - \dfrac{0.0591}{n}\log_{10} Q$

After calculation,

$E_{cell} = 1.41$ volt

49. 00000.60

Sol. Use MOT of O_2.

50. 00012.50

Sol. In a ccp lattice of oxide ions, there would be two tetrahedral and one octahedral void per oxide ion.

Out of the eight tetrahedral voids, one is occupied by A.

$\therefore$ % of tetrahedral voids occupied $= \dfrac{1}{8} \times 100 = 12.5\%$

Mathematics

PART – III

SECTION – A

51. C

Sol. α, β are roots of $x^2 - x - 3 = 0$

$\Rightarrow \alpha + \beta = 1, \alpha\beta = -3$

52. B

Sol. $x^3 + 3x^2 - 3x - 5 = (x + 1)(x^2 + 2x - 5)$

$x^3 + ax^2 + bx = x(x^2 + ax + b)$

Common root can be $-1 \Rightarrow a - b = 1$

or $a = 2, b = -5$

53. D

Sol. $p = {}^{1012}C_2, q = {}^{1011}C_2$

54. D

Sol. $f'(x) = 0 \Rightarrow f(x) =$ constant

$f(\alpha) + f(\beta) - 2f(\gamma) = 0$

55. C

Sol. $AB = A \Rightarrow A^2 = A, BA = B \Rightarrow B^2 = B$

$\Rightarrow (A+B)^{2020} = 2^{2019}A + 2^{2019}B = 2^{2019}(A+B)$

56. C

Sol. Let contestant choose chests A, B, C

x, y, z number of times respectively, then his chances of winning the prize is

$\frac{8}{13}\left[1-\frac{1}{2^x}\right]+\frac{4}{13}\left[1-\frac{1}{2^y}\right]+\frac{1}{13}\left[1-\frac{1}{2^z}\right]$ which is maximum for z = 672

57. B

Sol. Draw graph and see

58. D

Sol. $\frac{dy}{dt} = 3x^2\frac{dx}{dt} \Rightarrow 3x^2 \geq 10 \Rightarrow x \geq \sqrt{\frac{10}{3}}$

Distance of point $\left(\sqrt{\frac{10}{3}}, \left(\frac{10}{3}\right)^{\frac{3}{2}}\right)$ from origin is more than 6 units

59. A

Sol. Here, every critical point is point of extrema

If 'n' is odd, then $f'(x) = 0$ will be even degree polynomial and will have even number of real roots

60. A

Sol. $\int_0^n \{x\}f'(x)dx = \int_0^n xf'(x)dx - \sum_{r=1}^{n-1}\int_r^{r+1} rf'(x)dx = \int_0^n xf'(x)dx - \sum_{r=1}^{n-1} r\big(f(r+1)-f(r)\big)$

$$= \int_0^n x\, f'(x)dx + \sum_{r=1}^{n-1}\left(r f(r) - (r+1)f(r+1) + f(r+1)\right)$$

$$= \int_0^n x\, f'(x)dx + f(1) - nf(n) + f(2) + f(3) + ... + f(n)$$

$= -nf(n)$

61. C

Sol. $3y^2\frac{dy}{dx} - \frac{2}{x}y^3 = x$

Let $y^3 = t \Rightarrow \frac{dt}{dx} - \frac{2t}{x} = x$

$\Rightarrow \frac{y^3}{x^2} = \ln x + c$

62. B

Sol. '$\bar{z}$' is coplanar with $\bar{x}$ and $\bar{y}$

$\Rightarrow \bar{z} = \lambda\left(\frac{\bar{x}}{|\bar{x}|} + \frac{\bar{y}}{|\bar{y}|}\right)$

63. A

Sol. Ist and IIIrd are coincident planes and second is parallel to them so no solution

64. A

Sol. Normal to $y^2 = 4x$: $y = mx - 2m - m^3$ passes through (4, 0)

$\Rightarrow m = \pm\sqrt{2}$

$\Rightarrow$ Point on parabola $\left(2,\ 2\sqrt{2}\right)$

$\Rightarrow$ Shortest distance = $2\sqrt{3} - 1$

65. C

Sol. Let vertices be $\left(4\cos\theta_i,\ \sqrt{3}\sin\theta_i\right)$ where i = 1, 2, 3, then

Area of $\Delta = \frac{1}{2}4\sqrt{3}\begin{vmatrix}\cos\theta_1 & \sin\theta_1 & 1\\ \cos\theta_2 & \sin\theta_2 & 1\\ \cos\theta_3 & \sin\theta_3 & 1\end{vmatrix} = 2\sqrt{3}\left|\sin(\theta_2-\theta_1) + \sin(\theta_3-\theta_1) + \sin(\theta_2-\theta_3)\right|$

For maximum value $\theta_2 - \theta_1 = \frac{2\pi}{3}$, $\theta_3 - \theta_1 = \frac{\pi}{3}$ and $\theta_2 - \theta_1 = \frac{\pi}{3}$

$\Rightarrow$ Maximum area = 9

66. A

Sol. Equation of tangent having slope m : $y = mx \pm \sqrt{2m^2 - 1}$

$\Rightarrow 2 = \frac{2\sqrt{2m^2-1}}{\sqrt{1+m^2}} \Rightarrow m^2 = 2$. Intercepts are $\left(0,\ \pm\sqrt{3}\right)$ and $\left(\pm\frac{\sqrt{3}}{\sqrt{2}}, 0\right)$

67. A

Sol. $\tan 4x = \dfrac{4\tan x - 4\tan^3 x}{1 - 6\tan^2 x + \tan^4 x} = 1 \Rightarrow \tan^4 x + 4\tan^3 x - 6\tan^2 x - 4\tan x + 1 = 0$

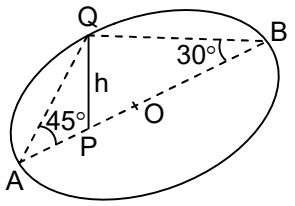
Harmonic mean = $\dfrac{4}{\dfrac{1}{\tan\alpha} + \dfrac{1}{\tan\beta} + \dfrac{1}{\tan\gamma} + \dfrac{1}{\tan\delta}} = 1$

68. B

Sol. $PB = \sqrt{3}\,h,\ AP = h$

$\Rightarrow OP = \left(\dfrac{\sqrt{3}-1}{2}\right)h$

$\angle OQP = \sin^{-1}\left(\dfrac{\sqrt{3}-1}{2}\right)$



69. D

Sol. See from truth table

70. D

Sol. Need to know the relative values of the 51 observation

SECTION – B

71. 1

Sol. $\lim\limits_{n\to\infty} \dfrac{1}{Q(n)} \sum\limits_{r=1}^{n} \dfrac{1}{\sqrt{a_r} + \sqrt{a_{r+1}}} = \lim\limits_{n\to\infty} \dfrac{n}{Q(n)\left(\sqrt{a_1} + \sqrt{a_{n+1}}\right)} = \lim\limits_{n\to\infty} \dfrac{n}{Q(n)\left(\sqrt{a_1} + \sqrt{a_1 + nd}\right)} = 0$

$\Rightarrow$ Degree of Q(n) is atleast 1

72. 4

Sol. X will be pair of lines both of which are parallel to AB and at a distance of $\dfrac{1}{2}$ units from it

SECTION – C

73. 01010.00

Sol. $z = e^{i\left(\frac{2n\pi + \frac{\pi}{2}}{2021}\right)}$ where n = 0, 1, ..., 2020

Re(z) > 0 for $n \in [0, 504] \cup [1516, 2020]$ $(n \in I)$

74. 00014.00

Sol. f(5) = 6, f(6) = 5 $\to$ 2

f(5) = 6, f(6) $\neq$ 5 $\to$ 4

f(5) $\neq$ 6, f(6) = 5 $\to$ 4

f(5) $\neq$ 6, f(6) $\neq$ 5 $\to$ 4

75. 00012.00

Sol. P is orthocentre of ΔBCD so line from P perpendicular to CD passes through B